

Proof. Assumptions. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ satisfy $X \geq 0$ a.s. Fix a constant $a > 0$.

Trivial cases.

1. If $\mathbb{E}[X] = +\infty$ then $\mathbb{E}[X]/a = +\infty$ and $\mathbb{P}(X \geq a) \leq +\infty$ holds automatically.
2. If $\mathbb{P}(X \geq a) = 0$ then the left-hand side of the desired inequality is 0, while the right-hand side is non-negative; thus the inequality holds with equality.

Hence we may restrict to the non-degenerate situation $\mathbb{E}[X] < \infty$ and $\mathbb{P}(X \geq a) > 0$.

1. Indicator function and pointwise bound. Define the measurable set

$$A := \{\omega \in \Omega : X(\omega) \geq a\}$$

and its indicator (characteristic) function

$$\mathbf{1}_A(\omega) = \mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Because $X \geq 0$ and $a > 0$, for every ω we have the pointwise inequality

$$\mathbf{1}_{\{X \geq a\}}(\omega) \leq \frac{X(\omega)}{a}. \quad (1)$$

Indeed,

- If $\mathbf{1}_{\{X \geq a\}}(\omega) = 0$ then $0 \leq X(\omega)/a$ since $X(\omega) \geq 0$.
- If $\mathbf{1}_{\{X \geq a\}}(\omega) = 1$ then $X(\omega) \geq a$ by definition of A ; dividing by the positive constant a yields $1 \leq X(\omega)/a$.

Thus (??) holds for all ω .

2. Taking expectations. Both sides of (??) are non-negative measurable functions, so the Monotonicity of Expectation applies:

Theorem (Monotonicity). *If Y, Z are random variables with $0 \leq Y(\omega) \leq Z(\omega)$ for all ω and $\mathbb{E}[Z]$ exists (finite or $+\infty$), then $\mathbb{E}[Y] \leq \mathbb{E}[Z]$.*

Applying the theorem with $Y = \mathbf{1}_{\{X \geq a\}}$ and $Z = X/a$ gives

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}\left[\frac{X}{a}\right]. \quad (2)$$

The left-hand side is, by definition of expectation of an indicator,

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (3)$$

The right-hand side simplifies by linearity of expectation (constants factor out):

$$\mathbb{E}\left[\frac{X}{a}\right] = \frac{1}{a}\mathbb{E}[X]. \quad (4)$$

Combining (??)–(??) yields Markov's inequality:

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}} \quad (5)$$

as claimed.

3. Equality case. Equality in (??) can occur only if equality holds in the monotonicity step (??). The monotonicity theorem states that $\mathbb{E}[Y] = \mathbb{E}[Z]$ holds *iff* $Y = Z$ almost surely. Hence we must have

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \frac{X(\omega)}{a} \quad \text{for } \mathbb{P}\text{-a.e. } \omega. \quad (6)$$

Equation (??) forces X to take only the two values 0 and a :

- If $X(\omega) < a$ then the indicator on the left is 0; (??) gives $X(\omega)/a = 0$, i.e. $X(\omega) = 0$.
- If $X(\omega) \geq a$ then the indicator equals 1; (??) gives $X(\omega)/a = 1$, i.e. $X(\omega) = a$.

Thus

$$X = a \mathbf{1}_{\{X \geq a\}} \quad \text{a.s.}$$

Equivalently, there exists $p \in [0, 1]$ such that

$$\mathbb{P}(X = a) = p, \quad \mathbb{P}(X = 0) = 1 - p,$$

and $p = \mathbb{P}(X \geq a)$. In this situation $\mathbb{E}[X] = ap$ and (??) becomes an equality:

$$\mathbb{P}(X \geq a) = p = \frac{ap}{a} = \frac{\mathbb{E}[X]}{a}.$$

Hence equality holds *iff* X is almost surely supported on the set $\{0, a\}$; otherwise the inequality is strict.

4. Conclusion. For every non-negative random variable X and every constant $a > 0$,

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}}$$

with equality precisely when X takes only the values 0 and a (with appropriate probabilities). The degenerate cases of infinite mean or zero probability have already been addressed, completing the proof. $\square$